AI Customer Churn Prediction System

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Submitted in partial fulfillment of the requirements for the degree of
Bachelor of Technology

in

Computer Science & AI
OR Computer Science & DS

by

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It is certified that the work contained in the project report entitled "AI Customer Churn Prediction System" by the following students has been carried out under my/our supervision and that this work has not been submitted elsewhere for a degree.

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19 April 2026

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Chapter 1

Introduction

Customer churn refers to the phenomenon where customers discontinue their subscription or stop using a company's services. In the telecom industry, churn rates typically range between 15% to 25% annually, making churn prediction a critical business problem.

Retaining existing customers is significantly less expensive than acquiring new customers. Therefore, predictive analytics techniques are widely used to identify customers who are likely to churn.

In this project, we first develop a machine learning-based churn prediction system using the Telco Customer Churn dataset. Baseline models such as Logistic Regression and Decision Tree are implemented to predict customer churn.

In the second phase, the system is extended into an intelligent agent-based application. This system not only predicts churn but also:

- Explains the reason behind churn predictions
- Retrieves relevant retention strategies
- Generates structured recommendations for business decisions

Thus, the project evolves from a predictive model into a complete AI-driven decision support system.

Chapter 2

Literature Review

Customer churn prediction has been widely studied using statistical and machine learning techniques. Traditional models such as Logistic Regression provide interpretable probabilistic outputs, making them suitable baseline models.

Tree-based models such as Decision Trees are capable of capturing non-linear relationships between features, improving predictive performance in many cases.

Recent advancements extend beyond prediction to include intelligent decision-support systems. The integration of Large Language Models (LLMs) enables systems to generate explanations and insights from model outputs.

Retrieval-Augmented Generation (RAG) further enhances these systems by allowing the use of external knowledge bases to provide context-aware recommendations.

Agent-based frameworks combine machine learning, LLM reasoning, and knowledge retrieval into a unified pipeline, enabling automated decision-making and strategy generation.

Thus, modern churn prediction systems are evolving from simple classification models to intelligent systems that support business decisions.

Chapter 3

Problem Formulation and Proposed Solution

3.1 Dataset Description

The dataset contains 7043 customer records with 21 features, including:

- Demographic information (Gender, Senior Citizen, Partner, Dependents)
- Service details (Internet, Phone, Streaming, etc.)
- Account information (Tenure, Contract type, Payment method)
- Billing details (Monthly Charges, Total Charges)

The target variable is **Churn**, which indicates whether a customer has discontinued the service.

3.2 Work Done So Far (Summary)

In the initial phase, data preprocessing and exploratory analysis were performed. The dataset was cleaned, transformed, and encoded for model training.

The following baseline models were implemented:

- Logistic Regression
- Decision Tree

These models were used to predict customer churn and establish a baseline for further system development.

3.3 Data Preprocessing

The following preprocessing steps were performed:

- Removal of irrelevant feature (customerID)
- Conversion of TotalCharges to numeric
- Removal of records with tenure = 0
- Handling missing values
- One-hot encoding of categorical variables
- Feature scaling using StandardScaler

After preprocessing, the final dataset consisted of 7032 records and 30 features.

3.3.1 Churn Distribution

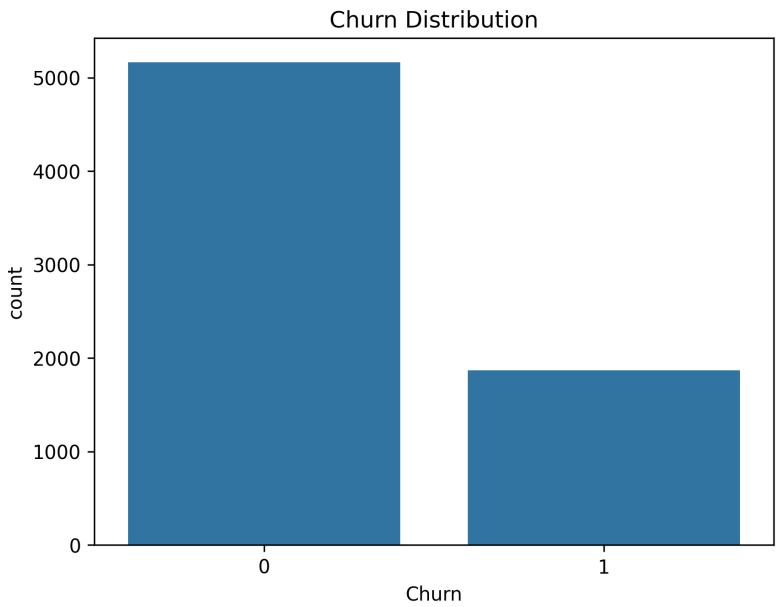

Figure 3.1: Distribution of Churn and Non-Churn Customers

3.4 Train-Test Split

The dataset was divided into:

- 80% Training data (5625 samples)
- 20% Testing data (1407 samples)

3.4.1 Correlation Analysis

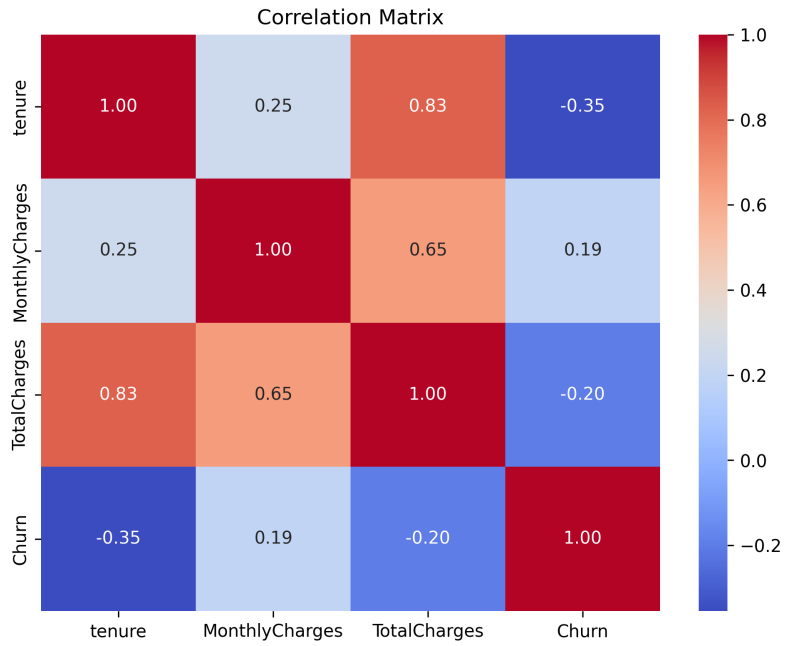

Figure 3.2: Correlation Matrix of Numerical Features

3.5 Baseline Models

3.5.1 Logistic Regression

Logistic Regression was used as a baseline classifier due to its simplicity and interpretability.

3.5.2 Decision Tree

A Decision Tree classifier was implemented to capture non-linear relationships within the dataset.

3.6 Proposed Agent-Based System

In Milestone 2, the system is extended into an intelligent agent-based application that not only predicts churn but also provides actionable insights and retention strategies.

3.6.1 Model Deployment and Prediction Pipeline

The trained Logistic Regression model, along with preprocessing components (scaler and feature columns), is loaded to ensure consistent predictions on new customer data.

A prediction function is designed to:

- Process input data
- Apply preprocessing steps
- Predict churn probability
- Classify risk level (High, Medium, Low)

3.6.2 LLM-Based Risk Reasoning

A Large Language Model (LLM) is integrated using Groq API to analyze customer data and prediction outputs.

The LLM:

- Interprets churn probability
- Identifies key risk factors
- Provides business-level explanations

3.6.3 Knowledge Base and Retrieval (RAG)

A knowledge base of retention strategies is created and stored as documents.

These documents are converted into embeddings and stored in a vector database (Chroma), enabling similarity-based retrieval.

Given a customer scenario, the system retrieves the most relevant retention strategies.

3.6.4 Agent Workflow using LangGraph

The complete system is structured as an agent workflow using LangGraph.

The workflow consists of the following steps:

- Prediction Node
- Reasoning Node

- Retrieval Node
- Report Generation Node

Each node processes and updates a shared state, forming a pipeline:

$$\text{Prediction} \rightarrow \text{Reasoning} \rightarrow \text{Retrieval} \rightarrow \text{Report}$$

3.6.5 Final Output

The system generates a structured report containing:

- Customer profile
- Churn probability and risk level
- AI-generated reasoning
- Recommended retention strategies

3.7 Mathematical Formulation of Logistic Regression

Logistic Regression models the probability of customer churn using the sigmoid function.

$$P(Y = 1|X) = \frac{1}{1 + e^{-z}}$$

Where:

$$z = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + ... + \beta_n x_n$$

The model is trained using binary cross-entropy loss:

$$L = -\frac{1}{m} \sum_{i=1}^{m} [y_i \log(p_i) + (1 - y_i) \log(1 - p_i)]$$

Chapter 4

Results and Discussion

4.1 Evaluation Metrics

The baseline models were evaluated using:

- Accuracy
- Precision
- Recall
- F1-score
- ROC-AUC score

4.2 Baseline Model Results

4.2.1 Logistic Regression

The Logistic Regression model achieved an accuracy of 78.7%.

- Accuracy: 78.75%
- Precision (Churn): 0.62
- Recall (Churn): 0.52
- F1-score (Churn): 0.56

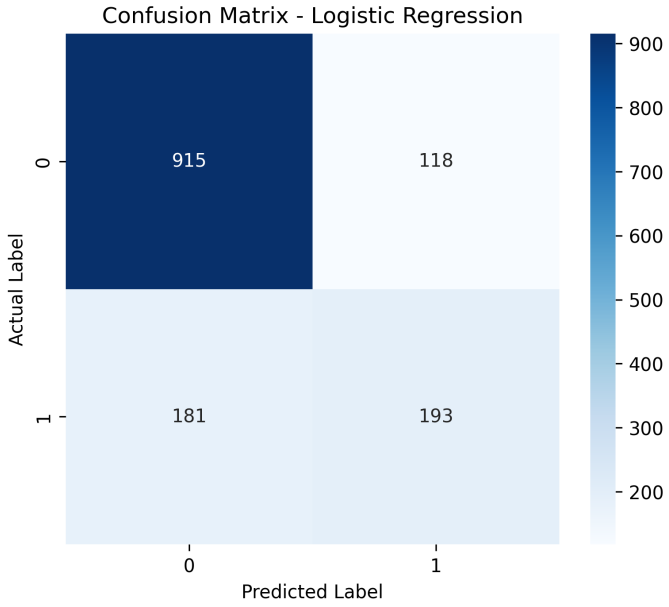

Figure 4.1: Confusion Matrix for Logistic Regression

4.2.2 Decision Tree

- Accuracy: 72.49%
- F1-score (Churn): 0.50

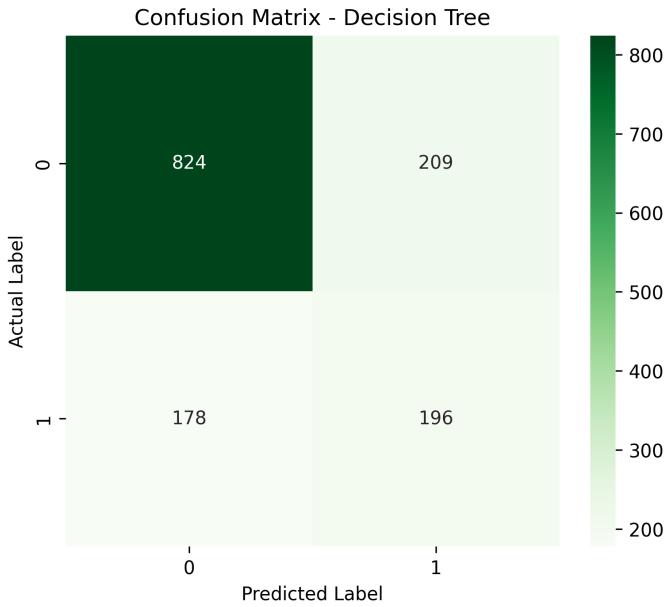

Figure 4.2: Confusion Matrix for Decision Tree

4.3 Agent-Based System Results

In Milestone 2, the system was extended into an AI agent capable of providing actionable insights along with churn prediction.

4.3.1 Prediction Output

The deployed model successfully predicts churn probability and assigns a risk level (High, Medium, Low) for new customer inputs.

4.3.2 LLM-Based Reasoning

The LLM generates explanations for churn risk by analyzing customer features and prediction results. These explanations provide business-level insights into why a customer is likely to churn.

4.3.3 Retrieval of Retention Strategies

Using a vector database, the system retrieves the most relevant retention strategies based on the customer profile and risk level.

4.3.4 Final Report Generation

The agent generates a structured report containing:

- Customer details
- Churn probability and risk level
- AI-generated reasoning
- Recommended retention strategies

4.4 System Analysis

The integration of machine learning with an agent-based framework significantly improves the usefulness of the system.

Instead of only predicting churn, the system now:

- Explains the prediction
- Suggests actionable steps
- Supports decision-making

4.5 Model Comparison

Model	Accuracy	F1-Score
Logistic Regression	78.75%	0.56
Decision Tree	72.49%	0.50

Table 4.1: Baseline Model Performance

Logistic Regression outperformed Decision Tree and was selected as the final model for deployment in the agent system.

4.6 Metric Definitions

Accuracy:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Precision:

$$Precision = \frac{TP}{TP + FP}$$

Recall:

$$Recall = \frac{TP}{TP + FN}$$

F1-Score:

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

Appendix A

Appendix

A.1 Appendix 1

A.2 Appendix 2

References